

Sheikh Rahat Mahmud

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[sheikh-rahata-mahmud](#) — 🌐 [Portfolio](#)

Education

Bangladesh University of Engineering & Technology (BUET), Dhaka **Expected Jun 2026**
Bachelor of Science in Computer Science and Engineering **Final Year (L-4 T-2)**
CGPA: 3.11 / 4.00 (Last Term(L-4T-1) GPA: **3.32 / 4.00**)

Higher Secondary Certificate (HSC) **2020 — GPA: 5.00 / 5.00**

Secondary School Certificate (SSC) **2018 — GPA: 5.00 / 5.00**

Relevant Coursework:

- **Core Systems:** Computer Networks, Info. System Design, Software Engineering, Microcontrollers & Embedded Systems, Computer Architecture, Operating Systems, Machine Learning, Bioinformatics.
- **Mathematical & Analytics:** Statistics, Discrete Math, Calculus, Linear Algebra, Economics.
- **Industrial & Business:** IPE (Industrial & Production Engineering) course, Accounting & Finance, Electronics, Electrical Machines, English course.

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, Java, C++, SQL
- **Frontend:** React.js, React Native, Material UI, HTML/CSS
- **Backend:** FastAPI, Django, Express.js, Node.js, REST APIs, Payment Gateway Integration
- **Databases:** PostgreSQL, Oracle, MongoDB
- **DevOps & Tools:** Git, CI/CD Pipelines, Docker, Linux, VM Deployment
- **Testing & QA:** Unit Testing, E2E Testing, UI Testing, CI/CD Pipeline Validation, Regression Testing
- **ML/DL & AI:** PyTorch, LangChain, LangGraph, CNN, Transformer, Knowledge Distillation
- **Model Architectures:** ResNet, UNet, 3D-UNet, Swin-UNet, UNETR, DNABERT-2, YOLOv8, YOLOv11
- **Other:** Agile Development, Code Review

Experience

Software Development Intern **Jan 2026 – Present**
CalcHVAC, New York (Remote)

- Delivering full-stack features using Python backend, React frontend, and Three.js for 3D HVAC simulation, with 10+ pull requests merged.
- Rendering interactive 3D simulation models with Three.js, enabling real-time design visualization for engineering workflows.

Intern **Nov 2025 – Dec 2025**
Bangladesh Satellite Company Limited (BSCL), Dhaka

- Automated database retrieval and reporting with Python scripts, processing 40,000+ records from Excel datasets for internal stakeholders.

Open Source Developer **Sep 2025 – Present**
Collaborating with [Shahad Ishraq](#) (Remote)

- Shipping 13 pull requests across 2 applications — a React Native mobile app and an Electron macOS desktop app.
- Maintaining WatermelonDB for offline-first data persistence, enabling full offline functionality with automatic sync on reconnect.
- Orchestrating agentic AI workflows with LangChain and LangGraph for multi-step reasoning and tool-based task execution across modular LLM agents.

Software Development Intern

Jun 2025 – Aug 2025

Academic & Admission Pathshala (EdTech), Dhaka

- Accelerated page load speed on 2 key pages (homepage and dashboard) through React component refactoring and lazy-loading optimization.

Projects

Thesis: Auto Annotation of 3D QCT Image Using Deep Learning

Jan 2026 – Present

Supervisor: Dr. Mahmuda Naznin, Professor, CSE, BUET

- Training 4 architectures (3D-UNet, UNet, Swin-UNet, UNETR) on a 22 GB+ DICOM dataset, achieving 97%+ accuracy for 3D QCT vertebral segmentation.

Lightweight Knowledge Distillation of Transformer Models for Biological Sequence Classification

Jan 2026 – Present

CSE-472 Machine Learning Project

- Compressing DNABERT-2 (117M parameters) into a 26M-parameter student model, retaining 93.31% accuracy (vs. 96.71% teacher) with 4.5x inference speedup.

IEEE Video & Image Processing (VIP) Cup

Jul 2025

Competition

- Benchmarked YOLOv8, YOLOv11, and TransNetV2 on a 2 GB+ dataset, achieving 90%+ accuracy in shot-boundary detection.

Career Studio– React.js, FastAPI, CI/CD, VM Deployment

Jan 2025 – Jul 2025

- Launched an AI-powered CV and portfolio builder with FastAPI backend and React.js frontend, deployed on an Azure VM.
- Implemented a CI/CD pipeline automating build, test, and release for continuous deployment to the Azure VM.

FaceBuet– React.js, Django, Oracle, Material UI

Mar 2023 – Jun 2024

- Built a social media platform for BUET students with 4 pages featuring posts, profiles, messaging, and search functionality using modular React.js components.
- Engineered a Django REST backend with Oracle database, implementing optimized SQL queries and search indexing across multiple tables.

MovieDatabase– Java, JavaFX, OOP Design Patterns

Jun 2022 – Aug 2022

- Created a JavaFX desktop application cataloging 300+ movie entries with search, filter, and analytics dashboards using OOP design patterns.

Certifications

- [CUET Datathon 2025 – Finalist](#)
- [BSCL Internship Completion Certificate](#)
- [Solvio AI Hackathon – Round 2 Qualifier](#)
- [CUET AI Hackathon 2025 – Participant](#) (Tested Whisper and Wav2Vec2 ASR models for Bangla speech recognition)